

Affiliate / Creator Code System

This document describes the affiliate/creator code system for Moody, allowing creators to earn commissions by referring new users.

Overview

The affiliate system enables:

- Creators to generate unique referral codes
- Users to apply referral codes at signup
- Automatic commission tracking on purchases
- Payouts via Stripe Connect

Architecture

Database Tables

- `affiliates` - Stores affiliate accounts and codes
- `user_referrals` - Links users to their referrer
- `affiliate_conversions` - Tracks purchases from referred users
- `affiliate_payouts` - Payout requests and history
- `referral_attempts` - Audit log for anti-fraud

Services

`affiliateService.ts`

Core affiliate logic:

- `createAffiliateCode(userId)` - Generate unique code for user
- `validateCode(code)` - Check if code exists and is active
- `applyCode(userId, code)` - Link user to affiliate
- `trackConversion(userId, amount, txId)` - Record purchase commission
- `getAffiliateStats(affiliateId)` - Dashboard statistics
- `getAffiliateEarnings(affiliateId)` - Earnings breakdown
- `requestPayout(affiliateId, amount)` - Request withdrawal

`payoutService.ts`

Stripe Connect integration:

- `createConnectedAccount(affiliateId, email)` - Onboard to Stripe
- `processPayoutRequest(payoutId)` - Execute payout transfer
- `getPayoutHistory(affiliateId)` - List past payouts
- Minimum payout threshold: \$20

React Hooks

`useAffiliate`

For affiliate creators:

```
const {
  isAffiliate,
  affiliate,
  stats,
  earnings,
  payoutHistory,
  stripeConnected,
  createCode,
  requestPayoutAmount,
  connectStripe,
} = useAffiliate();
```

useReferral

For referred users:

```
const {
  appliedReferral,
  pendingCode,
  applyReferralCode,
  storePendingCode,
  applyPendingCode,
} = useReferral();
```

Commission Structure

- Default commission rate: 10%
- Commission calculated on purchase amount
- Pending until purchase confirmed (not refunded)
- Available for payout after confirmation period

Anti-Fraud Measures

Rate Limiting

- Max 5 code applications per user per hour
- Prevents abuse of referral system

Self-Referral Prevention

- Users cannot apply their own affiliate code
- Checked at application time

Suspicious Activity Detection

- New account flagging (< 1 hour old)
- Multiple attempt tracking (> 10/day triggers block)
- Audit logging of all referral attempts

Payout Validation

- Minimum threshold: \$20
- Maximum single payout: \$10,000
- Balance verification before request

Stripe Connect Integration

Onboarding Flow

1. Creator clicks “Connect Stripe” in dashboard
2. Redirected to Stripe Express onboarding
3. Completes identity verification
4. Returns to app with connected account

Payout Flow

1. Creator requests payout from available balance
2. System creates payout record (pending)
3. Admin/cron processes payout
4. Stripe transfer executed to connected account
5. Status updated to completed/failed

Usage Examples

Creating an Affiliate Code

```
import { useAffiliate } from '@/hooks/useAffiliate';

function BecomeAffiliate() {
  const { createCode, loading } = useAffiliate();

  return (
    <Button onPress={createCode} disabled={loading}>
      Create My Code
    </Button>
  );
}
```

Applying a Referral Code

```
import { useReferral } from '@/hooks/useReferral';

function SignupScreen() {
  const { applyReferralCode } = useReferral();
  const [code, setCode] = useState('');

  const handleApply = async () => {
    try {
      await applyReferralCode(code);
      Alert.alert('Success', 'Referral code applied!');
    } catch (err) {
      Alert.alert('Error', err.message);
    }
  };
}
```

Tracking a Purchase

```
import { trackConversion } from '@services/affiliateService';

async function handlePurchaseComplete(userId, amount, transactionId) {
  // Track conversion for affiliate commission
  await trackConversion(userId, amount, transactionId);
}
```

Environment Variables

```
STRIPE_SECRET_KEY=sk_...
APP_URL=https://moody.app
```

Future Enhancements

- Tiered commission rates based on performance
- Custom codes (vanity codes)
- Referral link generation with UTM tracking
- Real-time conversion notifications
- Affiliate leaderboard
- Multi-currency support